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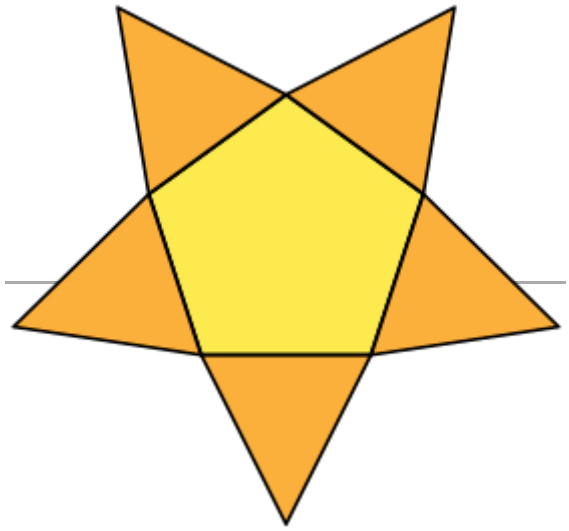
Pentagonal pyramid

In geometry, a **pentagonal pyramid** is a pyramid with a pentagonal base upon which are erected five triangular faces that meet at a point (the apex). Like any pyramid, it is self-dual.

The *regular* pentagonal pyramid has a base that is a regular pentagon and lateral faces that are equilateral triangles. It is one of the Johnson solids (J_2).

It can be seen as the "lid" of an icosahedron; the rest of the icosahedron forms a gyroelongated pentagonal pyramid, J_{11}.

More generally an order-2 vertex-uniform pentagonal pyramid can be defined with a regular pentagonal base and 5 isosceles triangle sides of any height.

Pentagonal pyramid	
Type	Johnson $J_1 - \boldsymbol{J_2} - J_3$
Faces	5 triangles 1 pentagon
Edges	10
Vertices	6
Vertex configuration	$5(3^2.5)$ (3^5)
Schläfli symbol	( ) ∨ {5}
Symmetry group	C_{5v}, [5], (*55)
Rotation group	C_5, $[5]^+$, (55)
Dual polyhedron	self
Properties	convex
Net	
	

Cartesian coordinates

The pentagonal pyramid can be seen as the "lid" of a regular icosahedron; the rest of the icosahedron forms a gyroelongated pentagonal pyramid, J_{11}. From the Cartesian coordinates of the icosahedron, Cartesian coordinates for a pentagonal pyramid with edge length 2 may be inferred as

$$(1,0,\tau),\ (-1,0,\tau),\ (0,\tau,1),\ (\tau,1,0),(\tau,-1,0),(0,-\tau,1)$$

where τ (sometimes written as φ) is the golden ratio.[1]

The height H, from the midpoint of the pentagonal face to the apex, of a pentagonal pyramid with edge length a may therefore be computed as:

$$H = \left(\sqrt{\frac{5-\sqrt{5}}{10}}\right) a \approx 0.52573a.$$ [2]

3D model of a pentagonal pyramid

Its surface area A can be computed as the area of the pentagonal base plus five times the area of one triangle:

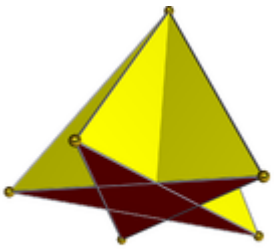
$$A = \frac{a^2}{2}\sqrt{\frac{5}{2}\left(10+\sqrt{5}+\sqrt{75+30\sqrt{5}}\right)} \approx 3.88554 \cdot a^2.$$ [3][2]

Its volume can be calculated as:

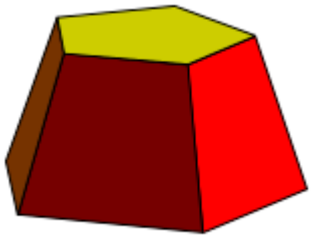
$$V = \left(\frac{5+\sqrt{5}}{24}\right) a^3 \approx 0.30150a^3.$$ [3]

Related polyhedra

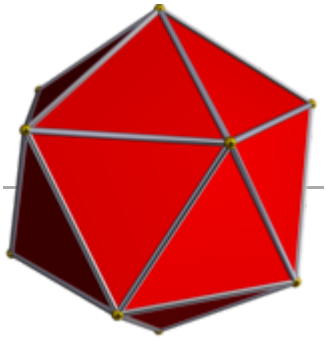
The pentagrammic star pyramid has the same vertex arrangement, but connected onto a pentagram base:



Regular pyramids [hide]						
Digonal	Triangular	Square	Pentagonal	Hexagonal	Heptagonal	...
Improper	Regular	Equilateral		Isosceles		
						...
						...

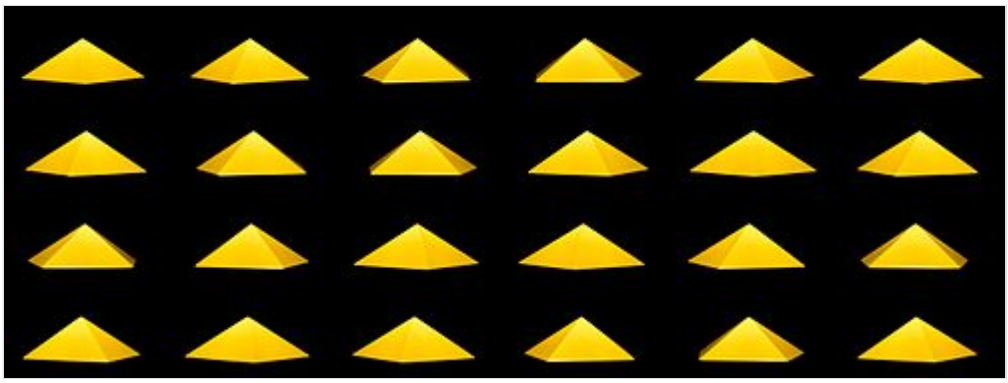


Pentagonal frustum is a pentagonal pyramid with its apex truncated



The top of an icosahedron is a pentagonal pyramid

Example



Pentagonal pyramid (at Matemateca IME-USP)

References

1. Weisstein, Eric W. "Icosahedral Group" (https://mathworld.wolfram.com/IcosahedralGroup.html). *mathworld.wolfram.com*. Retrieved 2020-04-12.
2. Sapiña, R. "Area and volume of a pentagonal pyramid and Johnson solid J_2" (https://www.problemasyecuaciones.com/geometria3D/volumen/Johnson/J2/calculadora-area-volumen-formulas.html). *Problemas y ecuaciones* (in Spanish). ISSN 2659-9899 (https://www.worldcat.org/issn/2659-9899). Retrieved 2020-06-29.
3. Weisstein, Eric W. "Pentagonal Pyramid" (https://mathworld.wolfram.com/PentagonalPyramid.html). *mathworld.wolfram.com*. Retrieved 2020-04-12.

External links

- Weisstein, Eric W., "Pentagonal pyramid (https://mathworld.wolfram.com/PentagonalPyramid.html)" ("Johnson solid (http://mathworld.wolfram.com/JohnsonSolid.html)") at *MathWorld*.
- Virtual Reality Polyhedra (http://www.georgehart.com/virtual-polyhedra/vp.html) www.georgehart.com: The Encyclopedia of Polyhedra (VRML model (http://www.georgehart.com/virtual-polyhedra/vrml/pentagonal_pyramid_(J2).wrl))

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